Albanese varieties and Albanese mappings Fall 2025

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The aim of this note is to give a self-contained introduction to Albanese varieties and Albanese mappings with view towards minimal model theory and classification of certain varieties. We will first give some elementary introduction to Albanese varieties and Albanese mapping. We will then dive into two research topic: Existence of good minimal for varieties with maximal Albanese dimension (Section 4), and Demailly–Peternell–Schneider's classification of compact Kähler manifolds with nef tangent bundle (Section 5). I will end this note by pointing out some further discussions.

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1 Albanese varieties

1.1 Construction over a compact Kähler manifold

Let us first construct the Albanese torus. The following observation is obvious: on a compact Kähler manifold holomorphic forms are automatically closed.

Lemma 1.1. Let X be a compact Kähler manifold with Kähler metric ω. If $\alpha \in H^0(X, \Omega_X^p)$ is a holomorphic p-form, then α is closed.

Proof. By definition $\bar{\partial}\alpha = 0$. By type reasons the formal adjoint satisfies $\bar{\partial}^*\alpha = 0$ as well, so $\Delta_{\bar{\partial}}\alpha = 0$. The Kähler identities give $\Delta_{\bar{\partial}} = \Delta_{\partial}$, so $\Delta_{\partial}\alpha = 0$. Then

$$0 = (\Delta_{\partial}\alpha, \alpha) = \|\partial\alpha\|^2 + \|\partial^*\alpha\|^2 \implies \partial\alpha = 0.$$

Together with $\bar{\partial}\alpha = 0$ this gives $d\alpha = 0$. □

The following proposition is important for the construction of Albanese torus for compact Kähler manifold.

Proposition 1.2. Let X be a compact Kähler manifold with Kähler form ω.
(1) There is a well-defined homomorphism

$$\varphi : H_1(X,\mathbb{Z}) \to (H^0(X,\Omega^1_X))^\vee, \qquad [\gamma] \mapsto \Big(\alpha \mapsto \int_\gamma \alpha\Big).$$

(2) The image of φ is a lattice in $(H^0(X,\Omega^1_X))^\vee$; hence the quotient is a complex torus of dimension $\dim_{\mathbb{C}} H^0(X,\Omega^1_X)$.

Proof of (1). By Lemma 1.1 every $\alpha \in H^0(X,\Omega^1_X)$ is closed. If $[\gamma] = [\gamma']$ in $H_1(X,\mathbb{Z})$, there is a singular 2-chain S with $\gamma - \gamma' = \partial S$, so by Stokes' theorem

$$\int_{\gamma-\gamma'} \alpha = \int_{\partial S} \alpha = \int_S d\alpha = 0.$$

Hence $\varphi([\gamma])$ depends only on the homology class $[\gamma]$. □

Proof of (2). Our proof follows [Huy05, page 135]. We must show that $H_1(X,\mathbb{Z})$ maps under φ to a lattice in $(H^0(X,\Omega^1_X))^\vee$. The natural map

$$H^{2n-1}(X,\mathbb{Z}) \to H^{2n-1}(X,\mathbb{C})$$

has image a lattice, since by the universal coefficient theorem tensoring with $\mathbb{Q}$ kills torsion. Applying Serre duality (SD) and Poincaré duality (PD) gives a commutative diagram

$$\begin{tikzcd} H^0(X,\Omega^1_X)^\vee & H^n(X,\Omega^{n-1}_X) \arrow[l, "\overset{\mathrm{SD}}{\simeq}"'] \\ H_1(X,\mathbb{Z}) \arrow[u, "\varphi"] \arrow[r, "\underset{\simeq}{\mathrm{PD}}"'] & H^{2n-1}(X,\mathbb{Z}) \arrow[u, "p"'] \end{tikzcd}$$

where p is the composite $H^{2n-1}(X,\mathbb{Z}) \to H^{2n-1}(X,\mathbb{C}) \to H^{n-1,n}(X)$, whose image is again a lattice (tensoring with $\mathbb{C}$ kills torsion).

The image of φ agrees with the image of p: by the definition of Poincaré duality, $\int_X \omega \wedge \mathrm{PD}([\gamma]) = \int_\gamma \omega$, while by the definition of Serre duality

$$\mathrm{SD}(\mathrm{PD}([\gamma]))(\omega) := \int_X \mathrm{PD}([\gamma]) \wedge \omega = \int_\gamma \omega = \varphi([\gamma])(\omega).$$

Hence $\mathrm{im}(\varphi) = \mathrm{im}(p)$ is a lattice, of full rank $2\dim_{\mathbb{C}} H^0(X,\Omega^1_X)$. □

Remark 1.3 (Lattices)**.** A lattice in a complex vector space V of dimension g is a discrete subgroup $\Lambda \subset V$ of maximal rank, i.e. a free abelian group of rank $2g$; the quotient V/Λ is then a complex torus of dimension g.

Definition 1.4 (Albanese torus, first version)**.** Let X be a compact Kähler manifold. The complex torus

$$\mathrm{Alb}(X) := \big(H^0(X,\Omega^1_X)\big)^\vee / H_1(X,\mathbb{Z})$$

is called the Albanese variety (or Albanese torus) of X; by Proposition 1.2 it is well defined. Fixing a base point $x_0 \in X$ produces the Albanese mapping (Definition 2.1 below).

2 The Albanese mapping

2.1 Construction of Albanese mapping

Definition 2.1 (Albanese mapping, [Huy05, Definition 3.3.7])**.** Let X be a compact Kähler manifold, fix $x_0 \in X$, and let $\omega_1, \dots, \omega_q$ be a basis of $H^0(X, \Omega_X^1)$. The Albanese mapping of X is

$$\mathrm{alb}_X : X \to \mathrm{Alb}(X), \qquad x \mapsto \Big(\int_{x_0}^{x} \omega_1, \dots, \int_{x_0}^{x} \omega_q \Big) \bmod H_1(X, \mathbb{Z}).$$

Proposition 2.2. The Albanese mapping is well defined, i.e., it does not depend on the choice of path from x_0 to x, and a different choice of base point translates alb_X by a constant.

Proof. Fix the basis $\omega_1, \dots, \omega_q$ of $H^0(X, \Omega_X^1)$ used to define alb_X. By Lemma 1.1, each ω_i is closed.

We first show that it is independence of the path. Let $x \in X$ and let γ, γ' be two paths from x_0 to x. Then $\sigma := \gamma \cdot (\gamma')^{-1}$ is a closed loop at x_0, and for each i,

$$\int_{\gamma} \omega_i - \int_{\gamma'} \omega_i = \int_{\gamma \cdot (\gamma')^{-1}} \omega_i = \int_{\sigma} \omega_i.$$

Hence

$$\Big(\int_{\gamma} \omega_1, \dots, \int_{\gamma} \omega_q \Big) - \Big(\int_{\gamma'} \omega_1, \dots, \int_{\gamma'} \omega_q \Big) = \varphi([\sigma]) \in \Lambda.$$

Since $\mathrm{alb}_X(x)$ is by definition the class of $\big(\int_{x_0}^{x} \omega_1, \dots, \int_{x_0}^{x} \omega_q \big)$ in $\mathbb{C}^q/\Lambda$, the two choices of path yield vectors in $\mathbb{C}^q$ differing by an element of Λ, hence define the *same* class in $\mathbb{C}^q/\Lambda$. Thus $\mathrm{alb}_X(x)$ does not depend on the path chosen, and alb_X is a well-defined map $X \to \mathrm{Alb}(X)$. $\square$

2.2 Universal property of Albanese mapping

Albanese mapping share a universal properties.

Theorem 2.3 (Universal property of the Albanese map; [Huy05])**.** Let $\varphi : X \to T$ be a holomorphic map from a compact complex manifold X (with base point x_0) to a complex torus T. There is a unique homomorphism of complex tori $\widetilde{\varphi} : \mathrm{Alb}(X) \to T$ such that

$$\varphi(x) = \widetilde{\varphi}(\mathrm{alb}_X(x)) + \varphi(x_0), \qquad x \in X.$$

$$\begin{tikzcd} X \arrow[r, "\mathrm{alb}_X"] \arrow[dr, "\varphi"'] & \mathrm{Alb}(X) \arrow[d, "\widetilde{\varphi}+\varphi(x_0)"] \\ & T \end{tikzcd}$$

Proof. Write $T = \mathbb{C}^g/\Lambda$ for a lattice $\Lambda \subset \mathbb{C}^g$. Let $\pi : \widetilde{X} \to X$ be the universal cover, with $\pi_1(X, x_0)$ acting as the deck transformation group, and fix a lift $\widetilde{x}_0 \in \widetilde{X}$ of x_0. Recall

$$\mathrm{Alb}(X) = H^0(X, \Omega_X^1)^{\vee} / H_1(X, \mathbb{Z}), \qquad \mathrm{alb}_X(x) = \Big(\omega \mapsto \int_{x_0}^{x} \omega \Big) \bmod H_1(X, \mathbb{Z}).$$

Since $\widetilde{X}$ is simply connected, the map $\varphi \circ \pi : \widetilde{X} \to T$ lifts along the universal cover $\mathbb{C}^g \to T$ to a holomorphic map

$$\phi = (\phi_1, \ldots, \phi_g) : \widetilde{X} \to \mathbb{C}^g, \qquad \phi(\widetilde{x}_0) = 0.$$

The lifting is equivaiant. For $\gamma \in \pi_1(X, x_0)$, both $\phi \circ \gamma$ and ϕ lift $\varphi \circ \pi$ (as $\pi \circ \gamma = \pi$), so they differ by a deck transformation of $\mathbb{C}^g \to T$:

$$\phi(\gamma \cdot z) - \phi(z) \in \Lambda \qquad \text{for all } z \in \widetilde{X}. \tag{$*$}$$

Fix γ and set $f_\gamma := \phi \circ \gamma - \phi$, a holomorphic map $\widetilde{X} \to \mathbb{C}^g$ valued in the discrete set Λ by ($*$). continuity of f_γ then forces f_γ to be locally constant. As $\widetilde{X}$ is connected, f_γ is a single constant $c_\gamma \in \Lambda$. So that differentiating,

$$\gamma^*(d\phi_i) = d\phi_i, \qquad i = 1, \ldots, g,\ \gamma \in \pi_1(X, x_0),$$

That is $d\phi_i$ is π_1-invariant holomorphic one form, so each $d\phi_i$ is the pullback $\pi^*\eta_i$ of a unique $\eta_i \in H^0(X, \Omega^1_X)$.

Fix a basis $\omega_1, \ldots, \omega_q$ of $H^0(X, \Omega^1_X)$, $q = h^0(X, \Omega^1_X)$, and write $\eta_i = \sum_{j=1}^q a_{ij}\omega_j$. The matrix $A = (a_{ij})$ defines a linear map

$$A : H^0(X, \Omega^1_X)^\vee \to \mathbb{C}^g, \qquad A(\xi)_i = \xi(\eta_i).$$

Every class in $H_1(X, \mathbb{Z})$ is represented by a loop γ at x_0, since $H_1(X, \mathbb{Z})$ is the abelianization of $\pi_1(X, x_0)$. Identifying $[\gamma]$ with the functional $\omega \mapsto \oint_\gamma \omega$, and lifting γ to a path from $\widetilde{x}_0$ to $\gamma \cdot \widetilde{x}_0$ in $\widetilde{X}$,

$$A([\gamma])_i = \oint_\gamma \eta_i = \int_{\widetilde{x}_0}^{\gamma \cdot \widetilde{x}_0} d\phi_i = \phi_i(\gamma \cdot \widetilde{x}_0) - \phi_i(\widetilde{x}_0) \overset{(*)}{\in} \Lambda.$$

Hence A descends to a homomorphism of complex tori

$$\widetilde{\varphi} : \mathrm{Alb}(X) = H^0(X, \Omega^1_X)^\vee / H_1(X, \mathbb{Z}) \longrightarrow \mathbb{C}^g/\Lambda = T.$$

For $x \in X$, lift a path from x_0 to x to a path in $\widetilde{X}$ from $\widetilde{x}_0$ to some $\widetilde{x}$ over x. Then

$$\varphi(x) - \varphi(x_0) = \phi(\widetilde{x}) - \phi(\widetilde{x}_0) \bmod \Lambda = \Big(\int_{x_0}^x \eta_1, \ldots, \int_{x_0}^x \eta_g \Big) \bmod \Lambda = A(\mathrm{alb}_X(x)) = \widetilde{\varphi}(\mathrm{alb}_X(x)),$$

using the fundamental theorem of calculus on each component and the definition of A. This gives

$$\varphi(x) = \widetilde{\varphi}(\mathrm{alb}_X(x)) + \varphi(x_0), \qquad x \in X.$$

□

2.3 Generator of albanese torus, projectivity of albanese varieties

Although the Albanese map is not surjective in general, the image of the albanese map however can generate the Albanese variety.

Proposition 2.4 ([Uen75]).

As a direct consequence of the previous proposition, we can immediate see that for Moishezon manifold, the Albanese torus is Abelian variety. For compact Kähler manifold, however the alabanese varieties needs not to be projective.

Proposition 2.5 ([Uen75, Proposition 9.15])**.** If X is a Moishezon manifold (in particular, if X is projective), then $\mathrm{Alb}(X)$ is an abelian variety.

Proof. There is an integer $n > 0$ such that the n-fold sum map

$$\mathrm{alb}_X^n : X \times \cdots \times X \longrightarrow \mathrm{Alb}(X), \qquad (z_1, \ldots, z_n) \mapsto \mathrm{alb}_X(z_1) + \cdots + \mathrm{alb}_X(z_n),$$

is surjective: the translates of $\mathrm{alb}_X(X)$ generate $\mathrm{Alb}(X)$ as a group (else, by the universal property, $\mathrm{Alb}(X)$ could be replaced by the strictly smaller subtorus that they do generate, contradicting minimality forced by uniqueness of the Albanese torus), and finitely many sums of a generating set already exhaust a connected compact group.

Since X is Moishezon, so is the product $X \times \cdots \times X$ (a finite product of Moishezon manifolds is Moishezon). The map alb_X^n is a proper surjective holomorphic map from this Moishezon manifold onto $\mathrm{Alb}(X)$, and a proper surjective holomorphic image of a Moishezon space is again Moishezon; hence $\mathrm{Alb}(X)$ is Moishezon.

Finally, a compact complex torus is Moishezon if and only if it is an abelian variety, since a compact complex torus is always Kähler and a Kähler Moishezon manifold is always projective. □

2.4 Extension to singular varieties

For applications to the minimal model program one needs the Albanese map for mildly singular (not necessarily smooth) varieties. This requires more than the linear algebra above.

Proposition 2.6 ([BS95, Lemma 2.4.1])**.** Let X be a complete normal variety in Fujiki's class $\mathscr{C}$ with at worst rational singularities. Then the Albanese map $X \dashrightarrow \mathrm{Alb}(X)$, defined via a resolution, is in fact everywhere-defined, i.e. a morphism $\mathrm{alb}_X : X \to \mathrm{Alb}(X)$.

Proof. Let $\mu : Y \to X$ be a resolution and set $\mathrm{Alb}(X) := \mathrm{Alb}(Y)$. Rational singularities give $H^1(X, \mathcal{O}_X) \cong H^1(Y, \mathcal{O}_Y)$ via μ^* (a Leray spectral sequence computation), from which one deduces that $\mu^* : \mathrm{Pic}^0(X) \to \mathrm{Pic}^0(Y)$ is étale (comparing tangent spaces at the identity) and injective (as μ is a birational morphism of normal varieties), hence an isomorphism. Dualising and invoking Theorem **??** identifies $\mathrm{Pic}^0(X)^\vee$ with $\mathrm{Alb}(Y)$, and the resulting map $X \to \mathrm{Pic}^0(X)^\vee$ — built from the Poincaré line bundle on $X \times \mathrm{Pic}^0(X)$ — is a genuine morphism extending $\mathrm{alb}_Y \circ \mu^{-1}$ across the singularities of X. See [BS95] and Reid's original treatment for the details. □

3 Irregularity and Albanese dimension

We now introduce the numerical invariants attached to $\mathrm{Alb}(X)$ and alb_X, following Ueno's book [Uen75], which is the standard reference for this material.

Definition 3.1 (Irregularity; [Uen75, Definition 9.20])**.** Let V be a complex variety with non-singular model V^*. The irregularity of V is

$$q(V) := \dim_{\mathbb{C}} H^1(V^*, \mathcal{O}_{V^*}),$$

independent of the choice of resolution V^* and a bimeromorphic invariant of V.

Definition 3.2 (Albanese dimension; [Uen75, Definition 9.21])**.** The Albanese dimension of a complex variety V is $t(V) := \dim \mathrm{Alb}(V^*)$, the dimension of the Albanese torus of a non-singular model V^*. We say V has maximal Albanese dimension if $t(V) = \dim V$, equivalently the Albanese map is generically finite onto its image.

Definition 3.3 (Augmented irregularity)**.**

$$\widehat{q}(X) := \sup\{\, q(X') \mid X' \to X \text{ a finite étale cover} \,\} \in \mathbb{Z}_{\geq 0} \cup \{\infty\}.$$

3.1 Comparison inequalities

Remark 3.4 ([Uen75, Corollary 9.8, Lemma 9.22])**.** In general the Albanese dimension is at most the irregularity, $t(V) \leq q(V)$, since $\mathrm{Alb}(V) \subseteq H^0(\Omega^1_V)^\vee$ by construction.

Proposition 3.5 (Behaviour under morphisms; [Uen75, Lemma 9.23])**.** Let $f : V \to W$ be a surjective morphism of complex varieties. Then

$$q(V) \geq q(W), \qquad t(V) \geq t(W).$$

Proof. Pullback of forms $f^* : H^0(W, \Omega^1_W) \to H^0(V, \Omega^1_V)$ is injective because f is dominant (a nonzero form cannot pull back to zero on a dense open subset where f is a submersion), which gives $q(V) \geq q(W)$. The Albanese-dimension inequality follows by applying this together with the universal property (Theorem 2.3): the composite $V \to W \to \mathrm{Alb}(W)$ factors through alb_V, so $\mathrm{Alb}(W)$ is (a quotient of) a further quotient torus of $\mathrm{Alb}(V)$. □

Lemma 3.6 ([Uen75, Lemma 9.11])**.** A meromorphic map from a compact complex manifold to a complex torus is automatically a morphism (holomorphic everywhere, with no points of indeterminacy).

Proposition 3.7 (Birational invariance; [Uen75, Proposition 9.12])**.** If $f : \widehat{X} \to X$ is a bimeromorphic morphism of complex manifolds, then $(\mathrm{Alb}(X), \mathrm{alb}_X \circ f)$ is the Albanese torus of $\widehat{X}$. Consequently, having maximal Albanese dimension is a birational invariant, and one may speak of the Albanese dimension of a singular or non-complete variety via any (equivalently, every) resolution.

Proof. Write $\alpha = \mathrm{alb}_X$, $\widehat{\alpha} = \mathrm{alb}_{\widehat{X}}$, and let $g : X \dashrightarrow \widehat{X}$ be the (in general only meromorphic) inverse of f. Fix $\widehat{x}_0 \in \widehat{X}$ and normalise $\alpha, \widehat{\alpha}$ so that $\widehat{\alpha}(\widehat{x}_0)$ and $\alpha(f(\widehat{x}_0))$ are the respective origins.

By Lemma 3.6, the meromorphic map $\alpha \circ f : \widehat{X} \to \mathrm{Alb}(X)$ is in fact a morphism, so the universal property (Theorem 2.3) produces a homomorphism of complex tori

$$A(f) : \mathrm{Alb}(\widehat{X}) \to \mathrm{Alb}(X), \qquad \alpha \circ f = A(f) \circ \widehat{\alpha}.$$

Symmetrically, $\widehat{\alpha} \circ g : X \dashrightarrow \mathrm{Alb}(\widehat{X})$ is, again by Lemma 3.6, a genuine morphism, giving a homomorphism

$$A(g) : \mathrm{Alb}(X) \to \mathrm{Alb}(\widehat{X}), \qquad \widehat{\alpha} \circ g = A(g) \circ \alpha.$$

Since $f \circ g = \mathrm{id}_X$ and $g \circ f = \mathrm{id}_{\widehat{X}}$ as bimeromorphic maps, substituting the two displayed identities into each other gives $A(f) \circ A(g) = \mathrm{id}$ and $A(g) \circ A(f) = \mathrm{id}$ on the dense open subsets where the relevant compositions are defined, and hence everywhere, since both sides are morphisms of complex tori and two morphisms of complex tori agreeing on a dense open subset agree everywhere. So $A(f)$ is an isomorphism, and $(\mathrm{Alb}(X), \alpha \circ f)$ satisfies the universal property of the Albanese torus for $\widehat{X}$. □

Lemma 3.8 (Leray–Blanchard subadditivity)**.** Let $g : X \to Y$ be a smooth fibration with connected fibres between compact Kähler manifolds, with fibre F. Then

$$q(X) \le q(Y) + q(F),$$

with equality if and only if the monodromy representation $\pi_1(Y) \to \mathrm{Aut}(H^1(F, \mathbb{Z}))$ is trivial.

Proof. Consider the Leray spectral sequence of the constant sheaf $\mathbb{R}_X$ for g,

$$E_2^{s,t} = H^s(Y, R^t g_* \mathbb{R}_X) \implies H^{s+t}(X, \mathbb{R}).$$

Since the fibres are connected and g is (differentiably) a locally trivial fibre bundle, $R^0 g_* \mathbb{R}_X = \mathbb{R}_Y$, and $R^1 g_* \mathbb{R}_X$ is a locally constant real vector bundle of rank $2q(F)$ containing $R^1 g_* \mathbb{Z}_X$ as a lattice, so its monodromy lies in $SL_{2q(F)}(\mathbb{Z})$; in particular

$$\dim H^0(Y, R^1 g_* \mathbb{R}_X) \le 2q(F),$$

with equality iff the monodromy is trivial. By Blanchard's theorem, the Leray spectral sequence of a Kähler fibration degenerates at E_2, so

$$\begin{aligned} 2q(X) &= \dim H^1(X, \mathbb{R}) \\ &= \dim H^1(Y, \mathbb{R}) + \dim H^0(Y, R^1 g_* \mathbb{R}_X) \\ &\le 2q(Y) + 2q(F), \end{aligned}$$

with equality iff the monodromy representation $\pi_1(Y) \to \mathrm{Aut}(H^1(F, \mathbb{Z}))$ is trivial. □

Lemma 3.9 (Descent to augmented irregularity)**.** In the situation of Lemma 3.8, $\widehat{q}(X) \le \widehat{q}(Y) + \widehat{q}(F)$.

Proof. The homotopy exact sequence of the fibration gives

$$\pi_2(Y) \to \pi_1(F) \to \pi_1(X) \to \pi_1(Y) \to 1.$$

A finite étale cover $\widetilde{X}$ of X corresponds to a finite-index subgroup $\pi_1(\widetilde{X}) \subset \pi_1(X)$; its image in $\pi_1(Y)$ and preimage in $\pi_1(F)$ are again finite-index, giving finite étale covers $\widetilde{Y}, \widetilde{F}$ fitting into a smooth fibration $\widetilde{X} \to \widetilde{Y}$ with fibre $\widetilde{F}$. By Lemma 3.8,

$$q(\widetilde{X}) \le q(\widetilde{Y}) + q(\widetilde{F}) \le \widehat{q}(Y) + \widehat{q}(F);$$

taking the supremum over $\widetilde{X}$ gives the claim. □

Lemma 3.10 (Rigidity of Kähler torus fibrations)**.** A smooth Kähler fibration by complex tori over a curve is holomorphically trivial.

Proof. Write the periods of a fibrewise Kähler class as

$$\mathrm{Im}({}^tPH\bar{P}) = Q,$$

where $P(z)$ is a holomorphic period matrix for the family and Q is a constant symplectic matrix (the periods of the integral lattice). A linear change of basis in Q reduces this, after writing P in block form and normalising one block to the identity, to a meromorphic matrix $B(z)$ satisfying ${}^tB = B$ and $\mathrm{Im}\, B = H^{-1}$ positive definite. Liouville's theorem then forces B, hence P, to be constant, i.e. the family of tori and their polarisations do not vary. □

Proposition 3.11 (Demailly–Peternell–Schneider's subadditivity; [**?**, Proposition 3.12])**.** Let $g : X \to Y$ be a smooth fibration with connected fibres between compact Kähler manifolds, with fibre F. Then

$$q(X) \le q(Y) + q(F), \qquad \widehat{q}(X) \le \widehat{q}(Y) + \widehat{q}(F).$$

If moreover Y is a complex torus and $\pi_1(F)$ is virtually abelian with the Albanese map of every finite étale cover of F of constant rank, then $\widehat{q}(X) = \widehat{q}(Y) + \widehat{q}(F)$.

Proof. The first two inequalities are Lemmas 3.8 and 3.9. For the equality statement, since $\pi_1(F)$ is finitely generated, the hypotheses give a free abelian subgroup $\Lambda \subset \pi_1(F)$ of finite index; take the finite étale cover $\widetilde{F}$ with $\pi_1(\widetilde{F}) = \Lambda$, so $q(\widetilde{F}) = \frac{1}{2}\mathrm{rk}\,\Lambda = \widehat{q}(F)$. The intersection

$$\Lambda' = \bigcap_{\varphi \in \mathrm{Aut}(\pi_1(F))} \varphi(\Lambda)$$

still has finite index (as Λ has only finitely many conjugates, each of bounded index) and is monodromy-invariant, giving a semidirect product $G' = \Lambda' \rtimes \pi_1(Y)$ mapping to $\pi_1(X)$ with finite-index image; the corresponding fibration $X' \to Y$ has fibre F' covering $\widetilde{F}$.

It remains to show the monodromy $\pi_1(Y) \to \mathrm{Aut}(H^1(F', \mathbb{Z}))$ has finite image: then some finite-index $\widetilde{H} \subset \pi_1(Y)$ acts trivially, giving a cover $\widetilde{X} \to \widetilde{Y}$ with fibre $\widetilde{F} = F'$ and trivial monodromy, so the equality case of Lemma 3.8 yields

$$\widehat{q}(X) \ge q(\widetilde{X}) = q(\widetilde{Y}) + q(\widetilde{F}) = \widehat{q}(Y) + \widehat{q}(F),$$

which with the reverse inequality already proved gives equality.

To bound the monodromy, define a Euclidean structure on $R^1 g_* \mathbb{R}_X$: for local sections u, v near $y \in Y$, representing classes in $H^1(F_y, \mathbb{R})$, Hodge theory writes $u = \mathrm{Re}(u')$, $v = \mathrm{Re}(v')$ for unique holomorphic 1-forms u', v' on F_y, and one sets

$$\langle u, v \rangle_y = \mathrm{Re} \int_{F_y} i\, u' \wedge \bar{v}' \wedge \omega^{p-1} \qquad (p = \dim F_y,\ \omega \text{ a Kähler form on } X).$$

If this structure is locally flat, the monodromy lies in $SL_{2q}(\mathbb{Z}) \cap O_{2q}(\langle\ ,\ \rangle)$, a finite group, as required.

Local flatness is checked via the relative Albanese factorisation $X \to A(X/Y) \to Y$ (which exists quite generally, and is easy to construct here since everything is smooth): $A(X/Y) \to Y$ is a smooth torus fibration with fibres $A(F_y)$. It is Kähler — fibre-by-fibre, the map

$$F^{\times N} \to A(F), \qquad (x_1, \dots, x_N) \mapsto \alpha(x_1) + \sum_{j \geq 2} (\alpha(x_j) - \alpha(x_1)),$$

is a submersion for $N \gg 0$ (its image's tangent spaces generate $T_{A(F)}$, since otherwise $\alpha(F)$ would lie in a proper subtorus, contradicting the universal property), so $A(X/Y)$ is a submersive image of the Kähler fibre product $X^{\times_Y N}$. Restricting to lines in Y and applying Lemma 3.10 gives local (indeed global, along each line) triviality of $A(X/Y) \to Y$, so every holomorphic 1-form on F_y extends to a d-closed holomorphic 1-form on a neighbourhood $g^{-1}(U)$; Stokes' theorem then shows $\langle u, v \rangle_y$ is independent of $y \in U$, i.e. the Euclidean structure is locally flat. □

Lemma 3.12 ([Uen75, Corollary 10.5])**.** If a proper subvariety B of a complex torus A generates A as a group (under translations), then $\kappa(B) > 0$.

Proof. If $\kappa(B) = 0$, then B is a translate of a complex subtorus of A. A translate of a proper subtorus cannot generate A, a contradiction. □

Proposition 3.13 ([Uen75, Corollary 10.6])**.** Let $\alpha : V \to \mathrm{Alb}(V)$ be the Albanese mapping. Then $\kappa(\alpha(V)) \geq 0$, with equality to 0 if and only if α is surjective.

Proof. Let $Z = \alpha(V) \subseteq \mathrm{Alb}(V)$. By the universal property, Z generates $\mathrm{Alb}(V)$ as a group (translates of $\alpha(V)$ additively span $\mathrm{Alb}(V)$, since $\mathrm{Alb}(V)$ was constructed as the dual of the forms pulled back from V). If α is not surjective, Z is a proper subvariety generating $\mathrm{Alb}(V)$, so $\kappa(Z) > 0$ by Lemma 3.12. If α is surjective, $Z = \mathrm{Alb}(V)$ has trivial canonical bundle, so $\kappa(Z) = 0$. In either case $\kappa(Z) \geq 0$, with equality precisely when α is surjective. □

4 Varieties with maximal Albanese dimension

One guiding principle organises this section: abelian varieties, and complex tori more generally, contain no rational curves. A variety mapping to a torus therefore has whatever rational-curve behaviour it possesses — exactly the behaviour the minimal model program acts on — confined entirely to the fibres of that map. Maximal Albanese dimension, where alb_X is generically finite so there is almost no fibre left, is therefore an unusually strong hypothesis, and interacts remarkably cleanly with the MMP (an observation due to Fujino).

4.1 A characterization via the cotangent bundle

Maximal Albanese dimension has an equivalent, purely infinitesimal formulation.

Proposition 4.1 ([**?**, Remark 2.2])**.** A smooth projective variety X has maximal Albanese dimension if and only if the evaluation map

$$H^0(X, \Omega^1_X) \otimes \mathcal{O}_X \longrightarrow \Omega^1_X$$

is surjective at the generic point of X, i.e. the cotangent bundle is generically globally generated.

Proof. The cotangent space of $\mathrm{Alb}(X)$ at any point is canonically $H^0(X, \Omega^1_X)$ (invariant 1-forms on a torus V/Λ are exactly $V^\vee$, so covectors at a point are $V = H^0(X, \Omega^1_X)^{\vee\vee} \cong H^0(X, \Omega^1_X)$). Unwinding Definition 2.1, the differential of alb_X at $x \in X$ is the evaluation pairing

$$d(\mathrm{alb}_X)_x : T_{X,x} \to H^0(X, \Omega^1_X)^\vee, \qquad v \mapsto (\omega \mapsto \omega(v)),$$

so its transpose $(d(\mathrm{alb}_X)_x)^\vee : H^0(X, \Omega^1_X) \to \Omega^1_{X,x}$ is exactly $\omega \mapsto \omega(x)$, i.e. the fibre at x of the evaluation map in the statement.

By Definition 3.2, X has maximal Albanese dimension exactly when alb_X is generically finite onto its image, i.e. when $d(\mathrm{alb}_X)_x$ is injective for generic x. Dualising an injective linear map between finite-dimensional spaces gives a surjective transpose, and conversely; so $d(\mathrm{alb}_X)_x$ is injective for generic x if and only if the evaluation map $H^0(X, \Omega^1_X) \to \Omega^1_{X,x}$ is surjective for generic x, which is the statement. □

4.2 Existence of good minimal models

Varieties of maximal Albanese dimension have unusually good minimal model theory, because the target of the Albanese map is (a subvariety of) an abelian variety, and abelian varieties contain no rational curves. The following lemma is the precise mechanism.

Lemma 4.2 ([**?**, Lemma 3.2, 3.3])**.** Let (X, Δ) be a log canonical pair and $f : X \to S$ a surjective morphism of projective varieties, with S containing no rational curves. If $K_X + \Delta$ is f-nef, then it is nef. Consequently, if (X', Δ') is a minimal model of (X, Δ) over S, it is already an (absolute) minimal model of (X, Δ).

Proof. If $K_X + \Delta$ were not nef, the cone theorem would produce a rational curve $C \subset X$ with $(K_X + \Delta) \cdot C < 0$. Since $K_X + \Delta$ is f-nef, C cannot be f-vertical, so $f(C)$ is a positive-dimensional subvariety of S; but $f(C)$ is a holomorphic image of $\mathbb{P}^1$, hence itself covered by rational curves, contradicting that S has none. The second statement is the same argument applied to $K_{X'} + \Delta'$, which is f-nef by definition of a relative minimal model. □

Remark 4.3. Abelian varieties, and more generally subvarieties of abelian varieties, contain no rational curves (a non-constant map $\mathbb{P}^1 \to A$ would lift to a non-constant holomorphic map $\mathbb{P}^1 \to \mathbb{C}^g$, impossible by compactness of $\mathbb{P}^1$). So Lemma 4.2 applies with S any subvariety of $\mathrm{Alb}(X)$ — in particular with $S = a_X(X)$.

Theorem 4.4 (Fujino; [**?**, Theorem 4.3])**.** Let (X, Δ) be a projective klt pair of maximal Albanese dimension. Then (X, Δ) has a good minimal model.

Proof. Klt singularities are in particular rational, so Proposition 2.6 shows $a_X : X \to \mathrm{Alb}(X)$ is an everywhere-defined morphism; let $S := a_X(X)$ and $f := a_X : X \to S$. Since X has maximal Albanese dimension, f is generically finite, hence $K_X + \Delta$ is f-big (a generically finite morphism makes every $\mathbb{R}$-divisor relatively big). By the relative existence theorem for minimal models of big divisors [BCHM10, Theorem 1.2], there is a minimal model $(X', \Delta') \to S$ of (X, Δ) over S.

Since $S \subseteq \mathrm{Alb}(X)$, Lemma 4.2 shows (X', Δ') is already an (absolute) minimal model of (X, Δ): $K_{X'} + \Delta'$ is nef. Finally, the abundance theorem for varieties of maximal Albanese dimension — a nef log canonical divisor on a projective klt pair of maximal Albanese dimension is automatically semiample, [?, Theorem 4.2] — upgrades this to $K_{X'} + \Delta'$ semiample: (X', Δ') is a good minimal model. □

Remark 4.5 (The Kähler case, due to Das–Hacon)**.** Fujino's argument leans on two purely projective/algebraic tools: the relative existence theorem for minimal models of big divisors [BCHM10], and the abundance theorem for maximal Albanese dimension varieties, both available only in the projective category. Das–Hacon [?] prove the same conclusion for compact Kähler klt pairs (X, B) of maximal Albanese dimension, but since neither tool is available as such in the Kähler category, the proof cannot simply run the argument above analytically; instead it replaces the abundance-theorem input with a direct argument via cohomology support loci and generic vanishing (Green–Lazarsfeld–Simpson GV-sheaves) to show the relevant divisors are a_X-exceptional, combined with an induction on the Iitaka fibration to handle $\kappa > 0$. This is substantially longer and uses heavier machinery than Fujino's projective proof, precisely because it has to reconstruct, by transcendental means, what BCHM and the abundance theorem supply for free in the projective case. Das–Hacon also record the corollary that if a_X happens to already be projective with connected fibres, one may run the $(K_X + B)$-minimal model program directly over $\mathrm{Alb}(X)$ (as in the proof of Theorem 4.4 above) to reach a good minimal model, without the transcendental argument.

5 Albanese fibrations for Kähler manifolds with nef tangent bundle

We now turn to Demailly–Peternell–Schneider's classification of compact Kähler manifolds with numerically effective (nef) tangent bundle, in which the Albanese fibration is again the central structural device — here playing the opposite role to Section 4: rather than being generically finite, alb_X turns out to be an honest smooth fibration, with the entire non-torus part of X concentrated, Fano and simply connected, in its fibres.

Theorem 5.1 (Demailly–Peternell–Schneider; [?, Theorem 3.14, 3.15])**.** Let X be a compact Kähler manifold with T_X nef, and let $\widetilde{X} \to X$ be a finite étale cover realising the maximal irregularity $q = \widehat{q}(X) = q(\widetilde{X})$. Then:

(i) $\pi_1(\widetilde{X}) \cong \mathbb{Z}^{2q}$;

(ii) the Albanese map $\mathrm{alb}_{\widetilde{X}} : \widetilde{X} \to \mathrm{Alb}(\widetilde{X})$ is a smooth fibration over a q-dimensional torus, with nef relative tangent bundle;

(iii) the fibres F of this fibration are Fano manifolds with nef tangent bundle.

Consequently $\pi_1(X)$ is, as an abstract group, an extension of a finite group by $\mathbb{Z}^{2q}$.

The proof is an induction on $\dim X$ that repeatedly feeds the theorem back into itself on the fibre F (of strictly smaller dimension); it uses two facts from the theory of nef tangent bundles as black-boxed inputs, both from [?]: that alb_X is always a submersion with nef relative tangent bundle when T_X is nef, and that a compact Kähler manifold with nef tangent bundle and vanishing augmented irregularity is automatically Fano. We also use the classical fact that Fano manifolds are simply connected (via Mori's theorem, Fano manifolds are covered by rational curves through every point, hence rationally connected, and rationally connected manifolds are simply connected).

Proof. We prove (i)–(iii) simultaneously by induction on $n = \dim X$; the case $n = 0$ is trivial.

The Albanese map is a smooth torus fibration. Pullback of a nef bundle is nef, so $T_{\widetilde{X}}$ is nef; by the cited structure result for manifolds with nef tangent bundle, $\mathrm{alb}_{\widetilde{X}}$ is a surjective submersion onto $\mathrm{Alb}(\widetilde{X})$, with connected fibres of nef relative tangent bundle. Since $\widetilde{X}$ was chosen with maximal irregularity, $\dim \mathrm{Alb}(\widetilde{X}) = q$. This is (ii).

The fundamental group of the fibre. Let F be a fibre of $\mathrm{alb}_{\widetilde{X}}$. Then T_F is nef (it is a quotient of $T_{\widetilde{X}}|_F$, restricted along the smooth fibration) and $\dim F < n$, so the inductive hypothesis for (i) applies to F: $\pi_1(F)$ is commensurable with $\mathbb{Z}^{2q(F)}$.

The fibre is Fano. Apply Proposition 3.11 in its equality case to the smooth fibration $\widetilde{X} \to \mathrm{Alb}(\widetilde{X})$ with fibre F: since $\widetilde{X}$ already realises $\widehat{q}(\widetilde{X}) = q = \widehat{q}(\mathrm{Alb}(\widetilde{X}))$, the equality $\widehat{q}(\widetilde{X}) = \widehat{q}(\mathrm{Alb}(\widetilde{X})) + \widehat{q}(F)$ forces $\widehat{q}(F) = 0$. Combined with T_F nef, the second black-boxed input from [**?**] now gives that F is Fano. This is (iii).

Fundamental group of $\widetilde{X}$. Since F is Fano, it is simply connected, so the long exact homotopy sequence of the fibration $F \to \widetilde{X} \to \mathrm{Alb}(\widetilde{X})$ collapses to an isomorphism $\pi_1(\widetilde{X}) \cong \pi_1(\mathrm{Alb}(\widetilde{X})) \cong \mathbb{Z}^{2q}$. This is (i).

Descending to X. Let $\widetilde{\widetilde{X}} \to \widetilde{X} \to X$ be the further finite étale cover obtained by intersecting all $\pi_1(X)$-conjugates of $\pi_1(\widetilde{X})$ inside $\pi_1(X)$; then $\widetilde{\widetilde{X}} \to X$ is Galois. Since $\pi_1(\widetilde{\widetilde{X}})$ has finite index in $\pi_1(\widetilde{X}) \cong \mathbb{Z}^{2q}$, it is itself $\cong \mathbb{Z}^{2q}$, and it sits inside $\pi_1(X)$ with finite (Galois) quotient. Hence $\pi_1(X)$ is an extension of the finite group $\pi_1(X)/\pi_1(\widetilde{\widetilde{X}})$ by $\mathbb{Z}^{2q}$. □

6 Further discussion

We close with brief pointers, in each case stated without proof, to results and open directions adjacent to the material above.

Remark 6.1 (Conditions for the Albanese map to be a fibration)**.** If the algebraic dimension $a(X) = 0$ (the classical invariant of the warning remark in §3), then alb_X has connected fibres [Uen75, Lemma 13.6]. If the image of alb_X is a curve C, then C is smooth of genus $\dim \mathrm{Alb}(X)$ with connected fibres [Uen75, Proposition 9.19]; and if $a(X) = 0$ with $t(X) = \dim X - 1$, then alb_X is an analytic fibre bundle outside a codimension-≥ 2 locus, with fibre an elliptic curve or $\mathbb{P}^1$ [Uen75, Theorem 13.8].

Remark 6.2 (A relative Albanese reduction, due to Ou)**.** Any fibre space $f : X \to Y$ between compact Kähler manifolds factors, after blow-ups, as $X \to W \to Y$, where the first map's fibres have extremal Albanese behaviour (vanishing irregularity, or maximal Albanese dimension) and the second map's fibres are non-uniruled; the construction repeatedly applies the *relative* Albanese reduction of the fibres of the current map, terminating because the relative Albanese dimension strictly decreases the residual fibre dimension at each step. This is the key technical tool in Ou's characterisation of uniruledness for compact Kähler manifolds [**?**, Proposition 8.6].

Remark 6.3 (Albanese fibrations and the abundance/Iitaka conjectures)**.** Using Kawamata's structure theorem for finite covers of abelian varieties [Kaw81] to reduce the Albanese map to a fibration over an abelian variety, Nakayama proved that $\kappa_\sigma(K_X + \Delta) = 0$ implies $\kappa(K_X + \Delta) = 0$ for klt pairs [Nak04, Corollary IV.4.9] — the case $\Delta = 0$ is the abundance conjecture in numerical Kodaira dimension 0. In a similar spirit, Fujino showed that the Iitaka fibration of a variety of maximal Albanese dimension has base again of maximal Albanese dimension, with fibres birational to an abelian variety isogenous to the kernel of the induced map on Albanese varieties [**?**, Proposition 2.4]. Hacon–Popa–Schnell proved the Iitaka conjecture $C_{n,m}$ outright whenever the base of a fibre space has maximal Albanese dimension [HPS18]; whether $C_{n,m}$ holds when only the *very general fibre* (rather than the base) has maximal Albanese dimension is a natural question, for which the relative Albanese reduction above is the expected tool.

Remark 6.4 (Good minimal models for Albanese-fibred varieties)**.** Complementing Theorem 4.4, Lai showed good minimal models exist for varieties fibred by pieces that themselves admit good minimal models; specialised to the Albanese map, if $a_X : X \to \mathrm{Alb}(X)$ is projective with connected fibres and the general fibre has a good minimal model, then so does X.

Remark 6.5 (Vanishing irregularity and rational connectedness)**.** Rationally connected varieties with rational singularities have $q(X) = 0$, indeed all their generalised plurigenera vanish. A partial converse holds for Kähler manifolds with pseudo-effective tangent bundle: there $\widehat{q}(X) \leq \dim X$ always, and $\widehat{q}(X) = 0$ forces X projective and rationally connected. Whether $q(X) = 0$ alone, with no positivity hypothesis on T_X, forces rational connectedness is open.

Remark 6.6 (Structure of varieties with nef anti-canonical divisor)**.** For X projective, or Kähler, with $-K_X$ nef, it is conjectured that alb_X is a locally constant fibration (étale-locally an isotrivial torus bundle); this generalises Theorem 5.1(i)–(ii) from nef tangent bundle to the strictly weaker nef anti-canonical hypothesis, and is known in increasing generality [Cao19, Wan22, CCM21, NW25, iMWWZ25, iMW23].

Remark 6.7 (The Beauville–Bogomolov–Yau decomposition)**.** For X compact Kähler with $c_1(X) = 0$, a finite étale cover splits as a product of a torus, Calabi–Yau factors, and holomorphic-symplectic factors (classically Beauville–Bogomolov in the projective case; extended to the Kähler klt setting by Bakker–Guenancia–Lehn [**?**]); the Albanese map realises exactly the projection onto the torus factor.

Remark 6.8 (Ueno's Conjecture K)**.** Ueno conjectured that when $\kappa(X) = 0$, the Albanese map is not merely a fibre space (as above) but becomes, after a finite étale base change $B \to \mathrm{Alb}(X)$, birationally trivial over B: $X \times_{\mathrm{Alb}(X)} B$ is birational to $F \times B$ over B, where F is the general fibre, with $\kappa(F) = 0$. This is open in general beyond the cases already covered by Theorem 5.1 and the Beauville–Bogomolov–Yau decomposition.

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